PROBLEM 4 | Problems T2_CrystStruct.pdf

Crystalline Structures & Miller Indices

Planar and Linear Densities in the BCC Lattice

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4. For a BCC structure, determine the surface (planar) density for the planes (100), (110) ***and*** (111)***; the linear density in the direction*** $[100]$***, and the volume density.*** *(Solution:* $\rho_{(100)} = 1/a^2$, $\rho_{(110)} = \sqrt{2}/a^2$, $\rho_{(111)} = 1/(a^2\sqrt{3})$, $\rho_{[100]} = 1/a$, $\rho_v = 2/a^3$*)*

Solution: $\rho_{(100)} = 1/a^2$, $\rho_{(110)} = \sqrt{2}/a^2$, $\rho_{(111)} = 1/(a^2\sqrt{3})$, $\rho_{[100]} = 1/a$, $\rho_v = 2/a^3$

PHASE 1: HYPOTHESIS & PARAMETERS

Physical Context:

A Body-Centred Cubic (BCC) lattice with lattice parameter a is studied.

BCC Lattice Data:

- Atoms per unit cell: $n = 2$ (8 at vertices $\times 1/8$ + 1 central).
- Volume of the cubic cell: $V_C = a^3$.
- Atomic radius: $R = \frac{a\sqrt{3}}{4}$ (contact along the cube diagonal $\langle 111 \rangle$).

PHASE 2: FUNDAMENTAL PHYSICS & FORMULA SELECTION

Applying the formal definitions of the course [Session 4 Slides 7-9]:

- **Planar Density ($\rho_{(hkl)}$):**

$$\rho_{(hkl)} = \frac{N_{\text{atoms contained in the face or section}}}{A_{(hkl)}}$$

- **Linear Density ($\rho_{[uvw]}$):**

$$\rho_{[uvw]} = \frac{N_{\text{atoms cut diametrically}}}{L_{[uvw]}}$$

- **Volumetric Density (ρ_v):**

$$\rho_v = \frac{n}{V_C} = \frac{2}{a^3}$$

PHASE 3: ANALYTICAL DERIVATION & CALCULATION STEP-BY-STEP

1. Planar Density on the (100) Plane:

- **Plane geometry:** Outer face of the cube, of dimensions $a \times a$.
- **Area:** $A_{(100)} = a^2$.
- **Atoms contained:** Only the 4 atoms located at the vertices of the square (the central atom of the BCC cell is at $z = a/2$, so it does not belong to the plane $z = 0$ or $x = 0$).

$$N_{\text{atoms}} = 4 \times \frac{1}{4} = 1 \text{ atom}$$

- **Planar density:**

$$\rho_{(100)} = \frac{1}{a^2} = \boxed{\frac{1}{\mathbf{a^2}}}$$

2. Planar Density on the (110) Plane:

- **Plane geometry:** Diagonal rectangle crossing the cube through two opposite edges.
- **Dimensions:** Base $= a\sqrt{2}$ (face diagonal), Height $= a$.
- **Area:**

$$A_{(110)} = a\sqrt{2} \times a = a^2\sqrt{2}$$

- **Atoms contained:**
- 4 vertices of the rectangle, each contributing $\frac{1}{4}$: $4 \times \frac{1}{4} = 1$ atom.
- 1 atom at the geometric centre of the cell (the (110) plane passes exactly through the centre of the cube): it contributes a whole 1 atom.
- Total atoms: $N_{\text{atoms}} = 1 + 1 = 2$ atoms.

- **Planar density:**

$$\rho_{(110)} = \frac{2}{a^2\sqrt{2}} = \frac{2\sqrt{2}}{2a^2} = \boxed{\frac{\sqrt{2}}{a^2}}$$

3. Planar Density on the (111) Plane:

- **Geometry and spacing:**

In the BCC lattice, the parallel (111) planes have an interplanar spacing $d_{111} = \frac{a}{\sqrt{1^2+1^2+1^2}} = \frac{a}{\sqrt{3}}$.

However, because the central atom sits exactly at the midpoint of the cube diagonal along $[111]$, the real (111) planes alternate at an effective distance $d_{\text{eff}} = \frac{a}{2\sqrt{3}}$ (corresponding to the reflected family $\{222\}$).

- **Direct calculation by cell or density relation:**

Using the relation $\rho_{(hkl)} = \rho_v \cdot d$ between the volume density and the spacing d of consecutive atomic layers (extension: the slides only define $\rho_{(hkl)}$ as atoms over area [Session 4 Slide 9]; the direct count below is the course method):

$$\rho_v = \frac{\rho_{(hkl)}}{d_{\text{eff}}} \implies \rho_{(111)} = \rho_v \cdot d_{\text{eff}} = \left(\frac{2}{a^3}\right) \cdot \left(\frac{a}{2\sqrt{3}}\right) = \boxed{\frac{1}{a^2\sqrt{3}}}$$

- **Direct check by geometric area:**

The plane that cuts the vertices $(a,0,0)$, $(0,a,0)$, $(0,0,a)$ forms an equilateral triangle of side $L = a\sqrt{2}$ and area $A = \frac{\sqrt{3}}{4}(a\sqrt{2})^2 = \frac{\sqrt{3}}{2}a^2$.

The plane does not contain the central atom (it lies at $(a/2, a/2, a/2)$, whose distance to the origin is $\frac{a\sqrt{3}}{2}$, while the plane is at $\frac{a}{\sqrt{3}}$).

Therefore, the plane contains only the 3 vertices: $N_{\text{atoms}} = 3 \times \frac{1}{6} = \frac{1}{2}$ atom.

$$\rho_{(111)} = \frac{1/2}{\frac{\sqrt{3}}{2}a^2} = \boxed{\frac{1}{a^2\sqrt{3}}}$$

4. Linear Density in the $[100]$ Direction:

- **Edge segment length:** $L_{[100]} = a$.
- **Atoms cut:** It cuts the two vertex atoms at their ends:

$$N_{\text{atoms}} = 2 \times \frac{1}{2} = 1 \text{ atom}$$

- **Linear density:**

$$\rho_{[100]} = \frac{1}{a} = \frac{\mathbf{1}}{\mathbf{a}}$$

5. Volumetric Density (ρ_v):

- It contains $n = 2$ atoms in a cell volume $V_C = a^3$:

$$\rho_v = \frac{\mathbf{2}}{\mathbf{a^3}}$$

PHASE 4: PHYSICAL INTERPRETATION & VERIFICATION

1. Ranking of the planes

Comparing the planar densities obtained in the BCC lattice:

- $\rho_{(110)} = \frac{\sqrt{2}}{a^2} \approx \frac{1.414}{a^2}$
- $\rho_{(100)} = \frac{1}{a^2} = \frac{1.000}{a^2}$
- $\rho_{(111)} = \frac{1}{a^2\sqrt{3}} = \frac{\sqrt{3}}{3a^2} \approx \frac{0.577}{a^2}$

$$\rho_{(110)} > \rho_{(100)} > \rho_{(111)}$$

The (110) plane is the densest plane of the BCC structure, which is why $\{110\}$ is the slip plane of BCC metals, with slip direction $\langle 111 \rangle$ [Session 4 Slide 37].

2. Cross-check with the volume density (extension)

Every plane of atoms carries $\rho_{(hkl)}$ atoms per unit area and consecutive atomic layers are a distance d apart, so a slab of area A and thickness d holds $\rho_{(hkl)} A$ atoms in a volume Ad: $\rho_v = \rho_{(hkl)}/d$. With $\rho_v = 2/a^3$:

Plane	Atomic layers (coordinates in units of a)	d	$\rho_v \cdot d$	Direct count
(100)	corners at $x = 0, 1$; body centre at $x = \frac{1}{2}$	$\frac{a}{2}$	$\frac{2}{a^3} \cdot \frac{a}{2} = \frac{1}{a^2}$	$\frac{1}{a^2}$
(110)	$x + y = 0, 1, 2$ (the body centre has $x + y = \frac{1}{2} + \frac{1}{2} = 1$)	$\frac{a}{\sqrt{2}}$	$\frac{2}{a^3} \cdot \frac{a}{\sqrt{2}} = \frac{\sqrt{2}}{a^2}$	$\frac{\sqrt{2}}{a^2}$
(111)	$x + y + z = 0, 1, \frac{3}{2}, 2, 3$	$\frac{a}{2\sqrt{3}}$	$\frac{2}{a^3} \cdot \frac{a}{2\sqrt{3}} = \frac{1}{a^2\sqrt{3}}$	$\frac{1}{a^2\sqrt{3}}$

For (111) the layers $x + y + z = 1$ and $\frac{3}{2}$ are $\frac{(3/2 - 1)\, a}{\sqrt{1^2 + 1^2 + 1^2}} = \frac{a}{2\sqrt{3}}$ apart (distance between parallel planes $x + y + z = c_1$ and c_2 is $|c_2 - c_1|\, a/\sqrt{3}$).

3. Linear density and contact

Along $[100]$ the atoms do not touch: their centres are a apart, while their diameter is $2R = \frac{a\sqrt{3}}{2} \approx 0.866\,a$ (contact is along $\langle 111 \rangle$ [Session 4 Slide 5]). Hence $\rho_{[100]} = 1/a$ is lower than the close-packed limit $1/(2R) = \frac{2}{a\sqrt{3}} \approx \frac{1.155}{a}$ that is reached along $[111]$.

4. Dimensional check

$[\rho_{(hkl)}] = \frac{\text{atoms}}{\text{m}^2}$, $[\rho_{[100]}] = \frac{\text{atoms}}{\text{m}}$ and $[\rho_v] = \frac{\text{atoms}}{\text{m}^3}$, consistent with a in m; in the cross-check, $[\rho_v \cdot d] = \text{m}^{-3} \cdot \text{m} = \text{m}^{-2}$.

5. Comparison with the official answer

All five results agree with the official solution: $\rho_{(100)} = 1/a^2$, $\rho_{(110)} = \sqrt{2}/a^2$, $\rho_{(111)} = 1/(a^2\sqrt{3})$, $\rho_{[100]} = 1/a$, $\rho_v = 2/a^3$.